MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE
MONOLITHIC 16-BIT MICROPROCESSOR, (FIXED INSTRUCTION)

This amendment forms a part of Military Specification MIL-M-38510/530(USAF), dated 10 November 1980, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1 and 1.2, delete and substitute:

"1.1 Scope. This specification covers the detail requirements for a monolithic N-channel, silicon gate microprocessor. Two product assurance classes are provided and are reflected in the complete part number.

"1.2 Part number. The part number shall be in accordance with MIL-M-38510."

* 1.3, delete and substitute the following entries:

"Maximum junction temperature at T_C = 125°C 1/- - - - - - - T_J = 150°C

"Thermal resistance, junction-to-case (θ_{JC}):
Case Q- (See MIL-M-38510, appendix C)"

* Bottom of page (before Beneficial comments), add the following footnote:

"1/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5009 of MIL-STD-883."

PAGE 2

* 3.5, delete and substitute:

"3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

* 3.7, delete and substitute:

"3.7 Marking. Marking shall be in accordance with MIL-M-38510."

* 3.8, second line: Delete "covering only NMOS microprocessors" and substitute "107".

PAGE 3

TABLE I, Test column, Input low voltage and Input high voltage: Delete "(except CLK)" and substitute "(except CLK, MN/$\overline{MX}$)" both places.

TABLE I, Test column, after Clock input high voltage, add two new tests:

"

Test	Symbol	Condition	Min	Max	Unit
MN/$\overline{MX}$ input low voltage	V_{IL3}	--	-0.5	0	V
MN/$\overline{MX}$ input high voltage	V_{IH3}	--	4.5	5.5	V

"

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* TABLE I, Test column, Output high voltage, Conditions column: Delete "I_{OH} = 400 A" and substitute "I_{OH} = -400 A".

TABLE I, Test column, Power supply current, Max limit column: Delete "340" and substitute "415".

PAGE 4

TABLE I, Test column, Ready setup time, Ready hold time, Request ($\overline{RQ}/\overline{GT}$) hold time, and Ready active to status passive: Delete these tests in their entirety.

PAGE 5

TABLE I, Test column, Address float to read active, $\overline{RD}$ inactive to next address active test: Delete these tests in their entirety.

TABLE 1, Min limits column: Delete all limits and substitute "--".

PAGE 6

TABLE I, Test column, $\overline{RD}$ width, $\overline{WR}$ width, ALE inactive to AD_0 -AD_{15} inactive, ALE inactive to $\overline{BHE}/S_7$, Address valid to ALE low, $\overline{BHE}/S_7$ valid to ALE low: Delete these tests in their entirety.

TABLE I, Test column, ALE active delay, ALE inactive delay, $\overline{WR}$, $\overline{INTA}$ active delay, $\overline{DT}/\overline{R}$, $M/\overline{IO}$ active delay, control inactive delay 1, Min limits column: Delete all limits and substitute "--".

PAGE 7

TABLE I, Test column, Control inactive delay 2, ALE inactive to AD_0 - AD_{15} inactive: Delete these tests in their entirety.

TABLE I, Test column, HLDA valid delay, $\overline{DEN}$ active delay, Address delay, Min limits column: Delete all limits and substitute "--".

PAGE 8

* TABLE II: Delete "Class C devices" column.

* TABLE II, MIL-STD-883 test requirements column, Group A test requirements: Delete "(Note 1)".

* TABLE II, delete "Additional electrical subgroups for group C periodic inspection" and associated entries.

* TABLE II, Group D end-point parameters, Class B devices: Delete "2,3,8" and substitute "2,8 (125°C)".

4.2a, delete and substitute:

"a. Burn-in test (method 1015 of MIL-STD-883).
1. Test condition D or E, using the circuit shown on figure 4, or equivalent.
2. T_A = 125°C minimum."

4.2c, first line, delete "test condition B" and substitute "test condition D".

4.2, add new item d. as follows:

"d. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer."

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PAGE 9

4.4.3c: Delete "condition B" and substitute "condition D".

4.4.4b: Delete "condition B" and substitute "condition D".

PAGE 13

* FIGURE 4, wiring diagram, pin 17 input designation: Delete "SP_1" and substitute "P_1". Designation of pin 19 input: Delete "P_1" and substitute "SP_1". Make these same change to pins 17 and 19 in the stress column of the pin/stress chart.

FIGURE 4, Waveform P_1: Delete "120 µs" (two places) and substitute "60 µs" (two places).

PAGE 22

TABLE IIIA, Symbols V_{IH1} and V_{IL1}, Terminal column: Delete "except CLK"(two places) and substitute "except CLK and MN/$\overline{MX}$" (two places).

TABLE IIIA, Symbol V_{IH2}, Input condition C column: Delete "4.0" and substitute "5.0".

TABLE IIIA, Symbol V_{IH2}, Input condition columns A, B, D thru I: Delete limits for each and substitute "4.5" (eight places).

TABLE IIIA, Symbol column, after V_{IL2}, add the following new tests:

Symbol	Pin number	Terminal	A	B	C	D	E	F	G	H	I	Figure
V_{IH3}	33	MN/$\overline{MX}$	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	--
V_{IL3}	33	MN/$\overline{MX}$	0	0	0	0	0	0	0	0	0	--

"

TABLE IIIA, Symbol column, t_{SLH1}, t_{HLH1}/t_{HHL1}, and t_{HLH2}: Delete these tests in their entirety.

TABLE IIIA, Symbol column, t_{SHL2} and t_{SLH3}/t_{SHL3}: Add footnote "1/".

Following table IIIA, add footnote:

"1/ Delete the conditions shown for tests 102 through 107A only."

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TABLE IIIB, Symbols V_{OH} and V_{OL}, Output condition A column: Delete "2.0" and "1.0" respectively, and substitute "1.5" both places.

TABLE IIIB, Symbol column, t_{PHL11}/t_{PLH11}: Add footnote "1/".

TABLE IIIB, Symbol column, t_{PLH24}: Delete in its entirety.

* Following table IIIB, add footnote:

"1/ Delete the conditions shown for test 102 through 107A only."

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* 4.6: Delete in its entirety.

* 6.1: Delete in its entirety.

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* 6.2, delete and substitute:

"6.2 Test vector table. Table V, test vector, which forms a part of this detail specification, is not printed herein due to its extreme length and complexity. Table V is a software control program of the sequence of vectors needed for driving the devices represented in this detail sheet as part of the inspection required by table II herein. Table V is only generated when a qualified source exists for this specification. When a qualified source exists, a single hard copy printout of table V is maintained for reference purposes only by the Defense Electronic Supply Center, DESC-ECS may be consulted for information on obtaining a copy."

* 6.4b., d., and g., delete and substitute:

"b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable."

"d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable."

"g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government."

PAGES 26 AND 27

TABLE IIIC, Test numbers 23 and 24: Delete in their entirety.

PAGES 30 and 31

* TABLE IIIC: Delete and substitute new table as printed on pages 6 and 7 of this amendment.

PAGES 32 AND 33

TABLE IIIC, Test numbers 108, 118, 125, 128, 129, 133, 134, 138, 140: Delete in their entirety.

TABLE IIIC, Test numbers 109 through 139, Min limits column for subgroups 9, 10, and 11: Delete all limits and substitute "--".

TABLE IIIC, Test number 111, Output under test column: Delete "A16/S_3 - A19/S_6" and substitute "A16/S_3 - A18/S_5".

PAGE 34

TABLE IIIC: Delete and substitute new table as printed on page 8 of this amendment.

PAGE 41

* 6.6, delete and substitute:

"6.6 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number."

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Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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TABLE IIIC. Group A inspection.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	Parameter verified	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				GND	AD_{14}	AD_{13}	AD_{12}	AD_{11}	AD_{10}	AD_9	AD_8	AD_7	AD_6	AD_5	AD_4	AD_3	AD_2	AD_1	AD_0	NMI	INTR	CLK	GND
1 T_C = 25°C	I_{CC}		101	Run pattern 3 of table IV using input condition C of table IIIA and V_{CC} = 5.5 V. Monitor I_{CC} throughout pattern to verify that it does not exceed limit, using output condition A of table IIIB																				
	V_{OL} V_{OH}	V_{OL}	102	Run pattern 2 of table IV using input condition C of table IIIA and V_{CC} = 5.5 V, and output condition B of table IIIB, with V_{LOAD} = 1.95 V (see figure 5)																			
	V_{OL} V_{OH}	V_{OH}	103	Run pattern 2 of table IV using input condition C of table IIIA and V_{CC} = 4.5 V, and output condition B of table IIIB, with V_{LOAD} = 2.1 V (see figure 5)																			
	V_{OL} V_{OH}	V_{IL1}	104	Run pattern 1 of table IV using input condition A of table IIIA and V_{CC} = 4.5 V, and output condition A of table IIIB.																			
	V_{OL} V_{OH}	V_{IL2}	105	Run pattern 1 of table IV using input condition E of table IIIA and V_{CC} = 5.5 V, and output condition A of table IIIB.																			
	V_{OL} V_{OH}	V_{IL3}	105A	Run pattern 1 of table IV using input condition C of table IIIA and V_{CC} = 4.5 V.																			
	V_{OL} V_{OH}	V_{IH1}	106	Run pattern 1 of table IV using input condition B of table IIIA and V_{CC} = 4.5 V, and output condition A of table IIIB.																			
	V_{OL} V_{OH}	V_{IH2}	107	Run pattern 1 of table IV using input condition D of table IIIA and V_{CC} = 5.5 V, and output condition A of table IIIB.																			
	V_{OL} V_{OH}	V_{IH3}	107A	Run pattern 1 of table IV using input condition C of table IIIA and V_{CC} = 5.5 V.																			
2	Repeat subgroup 1, tests 36-107A at T_C = +125°C, except limit for test 101 shall be asshown in limits columns.																						
3	Repeat subgroup 1, tests 36-107A at T_C = -55°C, except limit for test 101 shall be as shown in limits columns.																						
7	Run pattern 6 of table IV using input condition C of table IIIA and V_{CC} = 4.5 V and 5.5 V, and output condition A of table IIIB.																						
8	Repeat subgroup 7 at -55°C and +125°C.																						

See footnotes at end of table.

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TABLE IIIC. Group A inspection.
Terminal conditions (pins not designated are open).

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	RESET	READY	$\overline{TEST}$	$\overline{INTA}$ (QS1)	ALE (QS0)	$\overline{DEN}$ ($\overline{S0}$)	DT/$\overline{R}$ ($\overline{S1}$)	M/$\overline{IO}$ ($\overline{S2}$)	$\overline{WR}$ (LOCK)	HLDA (RQ/GT1)	HOLD (RQ/GT0)	$\overline{RD}$	MN/$\overline{MX}$	$\overline{BHE}$/S_7	A19/S_6	A18/S_5	A17/S_4	A16/S_3	AD_{15}	V_{CC}		Min	Max	
101																							340	mA
102																						--	0.45	V V
103																						2.4	--	V V
104																						2.4	0.45	V V
105																						2.4	0.45	V V
105A																						2.4	0.45	V V
106																						2.4	0.45	V V
107																						2.4	0.45	V V
107A																						2.4	0.45	V V
																						290		mA
																						415		mA

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TABLE IIIC. Group A inspection - Continued.

Symbol	Test number	Pattern number	Waveform figure	Parameter verified	Input cond. from table IIIA	Output cond. from table IIIB	Measurement terminal		Test limits 1/5/						Unit
							Input under test	Output under test	Subgroup 9 T_A = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
									Min	Max	Min	Max	Min	Max	
V_{OL} V_{OH}	142	6	7,8,9, 10,11	t_{PWH1}			CLK	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V
t_{PWH4}	143		7,8,9	---				ALE	t_{PWL1}-20	---	t_{PWL1}-20	---	t_{PWL1}-20	---	ns
V_{OL} V_{OH}	144	6	7,8,9,10, 11	t_{CYC}			CLK	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	145	6	7	t_{SHL1}	H	C,E 3/	READY	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	146		12	t_{SHL2}	I	D,F 3/	$\overline{RQ/GT0}$ $\overline{RQ/GT1}$	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	147		13	t_{SHL3} t_{SLH3}	H	C,E 3/	HOLD	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	148		14	t_{SHL4} t_{SLH4}	H	C,E 3/	NMI INTR $\overline{TEST}$	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	149		7,11	t_{SLH1}			READY	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	150		8,10	t_{SZH} t_{SZL}	H	C,E 3/	AD_0-AD_{15}	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	151		7,11	t_{HHL1} t_{HLH1}			READY	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	152		12	t_{HLH2}			$\overline{RQ/GT0}$ $\overline{RQ/GT1}$	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V
V_{OL} V_{OH}	153		8,10	t_{HHZ} t_{HLZ}	H	C,E 3/	AD_0-AD_{15}	All outputs	1.5	1.5	1.5	1.5	1.5	1.5	V V

1/ Subgroups 9, 10, and 11 are performed with V_{CC} = 4.5 V and 5.5 V.

2/ Subgroup 9, patterns 7-15 are applied with input condition F from table IIIA. Patterns 16-44 are applied with input condition G from table IIIA.

3/ The capacitance measurements shall be made between the indicated terminal and ground at a frequency of 1 MHz. The dc bias of the measuring instrument shall be less than ±0.1 V. The ac signal amplitude shall be less than 50 mVrms.

4/ Minimum delay times tested during initial qualification and after process or design changes.

5/ Where propagation delay tests for subgroup 9 indicate a group of pins and where two or more propagation delays are specified for a test, only the reading for the pin with the worst case delay requires recording. In some cases this will be a minimum reading while for others it will be a maximum reading as indicated in the table.

6/ Clock is pulsed continuously between 0.0 and 4.0 V.

7/ This delay is for outputs AD_0-AD_{15}, A19/S_6-A16/S_3, $\overline{S_0}$-$\overline{S_2}$, $\overline{RD}$, $\overline{LOCK}$, and $\overline{BHE}$/S_7 during request/grant and HOLD/HOLDA sequences only.

8/ Output conditions C and D are used for subgroup 9, E and F are used for subgroups 10 and 11.

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